

CYBER DETECTIVE 2026

Forensic Investigation & Cyber Puzzle Competition Platform

PRODUCT REQUIREMENTS DOCUMENT

Document	Product Requirements Document (PRD)
Product	Cyber Detective 2026
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Status	Approved for build
Platform	Web (Flask + SQLite, localhost deployment)
Administrator	Competition organizers / platform admin
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Owner	Product Owner, Forensics Case Team

Table of Contents

1. Executive Summary
2. Background & Problem Statement
3. Goals & Objectives
4. Scope
5. Target Users & Personas
6. Functional Requirements
7. Non-Functional Requirements
8. Data & Information Architecture
9. UI / UX Requirements
10. Success Metrics & KPIs
11. Milestones & Timeline
12. Risks & Mitigations
13. Acceptance Criteria
14. Glossary

1. Executive Summary

Cyber Detective 2026 is a two-round technical competition platform that teams play in a live, timed environment. Round 1 (*Mixed Fundamentals*) delivers variant-based cyber puzzle challenges with a staged virtual lab and a hint economy. Round 2 (*Digital Forensics*) immerses teams in narrative case investigations involving persons of interest, digital evidence artifacts, and a final written report.

The platform is fully self-service for organizers: a built-in Admin Control Center provisions teams, controls round availability, curates content, monitors live activity, audits administrator actions, and publishes leaderboards. The product runs as a single local Flask application on `127.0.0.1:5000` with an embedded SQLite database — no external services required.

2. Background & Problem Statement

Running a hands-on cyber competition typically requires a patchwork of tools: one system for team registration, another for challenges, another for forensic case packets, plus manual scorekeeping. Organizers lose time, make transcription errors, and cannot respond to live conditions (pausing rounds, adjusting attempts, freezing teams).

Problem statement: Organizers need a single, controlled, offline-ready system that admits teams with unique round-specific credentials, delivers both puzzle and investigation content, enforces round policies (timers, attempt limits, access control), and gives administrators complete operational visibility — all without external dependencies.

3. Goals & Objectives

#	Goal	Success evidence
G1	One self-contained deployment	Runs from a single Python process + SQLite file
G2	Secure, admin-controlled access	Admin-provisioned team IDs per round; hashed admin passwords
G3	Full content lifecycle management	CRUD + preview for all Round 1 & Round 2 content
G4	Live operational control	Round status toggles, team freeze, session clearing
G5	Transparent scoring & reporting	Leaderboards, submission review, CSV exports, audit log

4. Scope

4.1 In Scope

- Participant journeys for both rounds (login, dashboards, challenges, labs, cases, evidence, reports, scores, leaderboards).
- Admin Control Center: teams, round settings, Round 1 challenges/variants/validation/ scoring, Round 2 cases/stations/persons/evidence, monitoring, audit, reports.
- Evidence artifact serving and downloads.
- Team & round access enforcement, timers, and maximum-attempt logic.

4.2 Out of Scope (future)

- Public internet deployment, SSO, and multi-tenant hosting.
- Real-time multiplayer chat or remote proctoring.
- Mobile native applications (the UI is responsive web only).
- Anti-cheat beyond attempt/hint/session instrumentation.

5. Target Users & Personas

Persona	Goals	Pain points addressed
Competition Admin	Provision teams, run the event, monitor live progress, adjust content	No single pane of glass; manual credential spreadsheets
Participant (Team Member)	Sign in with simple credentials, complete challenges, submit answers, track score	Fragmented logins per round; unclear rules/attempts
Content Author	Add/edit challenges, cases, and evidence; preview before publishing	No safe editing path; risky raw DB edits

6. Functional Requirements

FR1 — Access & Authentication (high)

- **FR1.1** Participants log in per round using **team name + round-specific access ID** issued by the admin.
- **FR1.2** Admin authenticates with username/password (hashed via Werkzeug).
- **FR1.3** Access is enforced per round (`access_enabled`) and per team (`is_active` , R1/R2 access IDs).
- **FR1.4** Round status gates login: `DRAFT / SCHEDULED / ACTIVE / PAUSED / COMPLETED` .
- **FR1.5** Pausing or disabling a round/team invalidates the participant session and redirects to the login page with a notice.
- **FR1.6** Admins can clear active sessions individually, in bulk, or globally.

FR2 — Round 1: Cyber Puzzles (high)

- **FR2.1** R1 challenges have code, domain (WEB/CRYPTO/FORENSICS/NETWORK/OSINT), difficulty, points, ordering, and an active flag.
- **FR2.2** Each challenge holds multiple *variants* (question bank) with a validation mode and expected answer.
- **FR2.3** Variants support validation modes: `NORMALIZED` , `EXACT` , `CONTAINS` , `CASE_SENSITIVE` .
- **FR2.4** Teams receive assignments; per-lab **max attempts** come from round settings and are enforced live.
- **FR2.5** Virtual labs stage sequential actions and log events; teams may request hints (tracked), then submit answers.
- **FR2.6** A timed session (timer from settings) tracks completion; results flow to the leaderboard.

FR3 — Round 2: Digital Forensics (high)

- **FR3.1** Cases are authored with title, code, company, brief, difficulty, status, and ordering; only ACTIVE cases are playable.
- **FR3.2** Each case contains investigation task stations (name, domain, points, max attempts, validation mode, description, question, answer, hint).
- **FR3.3** A person directory (role/key/answers/clue) supports interviews.
- **FR3.4** Evidence items reference artifact files (from the `evidence/` folder) that participants download.
- **FR3.5** Teams submit a final report and their score reflects solved stations and overall case progress.

FR4 — Scoring & Leaderboards (medium)

- **FR4.1** Admin-configurable scoring modes per round.
- **FR4.2** Correct submissions count toward team scores.
- **FR4.3** Leaderboards render by round with participant-visible scoreboard.
- **FR4.4** Admin review screens show submissions with acceptance status.

FR5 — Admin Control Center (high)

- **FR5.1** Overview dashboard with live aggregate stats.
- **FR5.2** Team management: create, edit, import (CSV), export, enable/disable, round access IDs, progress reset.
- **FR5.3** Full CRUD for every content entity with **prefill API endpoints** so edit modals never show blank/ghost values.
- **FR5.4** Live monitoring of sessions, audit log of admin actions, submissions review, and CSV/system exports.
- **FR5.5** Admin UI is a fixed on-screen frame: locked sidebar/topbar with internal content scrolling and full keyboard (Esc) modal handling.

FR6 — Ops & Reporting (medium)

- **FR6.1** CSV exports for teams, assignments, events, and submissions.
- **FR6.2** Audit log records label + target for admin mutations.
- **FR6.3** Case-code counter for Round 2 maintains unique case codes.

7. Non-Functional Requirements

Category	Requirement
Performance	Synchronous page loads < 300 ms for localhost usage; JSON APIs < 100 ms.
Usability	Keyboard-friendly modals (Esc to close); consistent dark admin theme; responsive to narrow viewports.
Reliability	Single-process, single-port deployment with no external services; crash-free for event duration.
Security	Hashed admin passwords; session-based auth; localhost-only binding; audit trail; no secrets committed.
Maintainability	Modular Python (app / access / admin_ops / round1 package); CSS custom-property design system.

Compatibility	Chrome/Safari/Firefox current versions; SQLite via Python stdlib.
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8. Data & Information Architecture

Single SQLite database `round1/round1.db` :

Entity	Description
<code>teams</code>	Authorized teams, R1/R2 access IDs, active flag
<code>admins</code>	Admin accounts with hashed passwords
<code>round_settings</code>	Timer, max attempts, access & status per round
<code>challenge_categories</code>	R1 challenges catalog
<code>challenge_variants</code>	R1 question variants with validation config
<code>team_challenge_assignments</code>	Per-team R1 assignments & progress
<code>participant_sessions</code> / <code>round_sessions</code>	Session lifecycle
<code>submissions</code>	Answer submissions & correctness
<code>hint_usage</code> / <code>lab_events</code>	Instrumentation trails
<code>r2_cases</code> / <code>r2_stations</code> / <code>r2_persons</code> / <code>r2_evidence</code>	Round 2 case content
<code>admin_audit_log</code>	Admin action history

9. UI / UX Requirements

- Purple/dark identity with locked app frame for Admin; light, lesson-style frames for participants.
- Status semantics: success green `#00E6A8` , warning amber `#FFB020` , danger `#FF4D6D` , accent purple `#6C2BFF` .
- Every destructive or bulk action confirms before executing and shows an in-app toast.
- Edit dialogs must prefill from dedicated API endpoints; never rerender stale row data.
- Empty states, loading toasts, and keyboard-accessible modal workflows are required.

10. Success Metrics & KPIs

Metric	Target
Smooth event operation	0 admin interventions required for content/access issues
Zero data loss	All submissions, hints, and scores persisted
Content authoring speed	Add/edit a challenge or case in < 3 minutes
Adoption	100% of registered teams complete at least one round

11. Milestones & Timeline

Phase	Deliverable	Status
M1 — Foundation	Flask app, SQLite schema, auth, Round 1 engine	Done
M2 — Round 2	Cases, stations, persons, evidence, reports	Done
M3 — Admin Control Center	Full CRUD, monitoring, audit, scoring, previews	Done
M4 — Hardening	Layout fixes, modal reliability, cache-busting	Done
M5 — Release	Git push, README, PRD packaging	In progress

12. Risks & Mitigations

Risk	Impact	Mitigation
In-process template caching hides changes	Admin sees stale UI	Documented restart procedure; CSS cache-busting <code>?v=</code>
Blank edit modals	Content corruption	Dedicated prefill API endpoints + server restarts
SQLite single-user writes	Contention under load	Localhost-only competition scale; short transactions
Fixed secret key	Session forgery in shared envs	Rotate via environment before external deployment

13. Acceptance Criteria

- Admin can provision a team that signs in to both rounds with round-specific IDs.
- Round pause immediately prevents new sign-in and funnels existing users to a notice.
- Every Round 1 variant and Round 2 question validates answers with the configured mode.
- Every edit dialog opens with correct prefilled data and saves without data loss.
- The entire admin application fits the screen with internal content scrolling.
- All 25+ admin pages render successfully and return HTTP 200 to an authenticated admin.

14. Glossary

Term	Definition
Variant	A question/answer instance of a Round 1 challenge
Station	An investigation task inside a Round 2 case
Access ID	Admin-issued per-round credential for a team
Validation mode	Answer comparison rule (normalized, exact, contains, case-sensitive)
Preview	Admin-only render of participant-facing content without answer keys